

Tutorial 1 – Advanced Array Manipulation

One Dimensional Array

1. Write a C++ program to perform the following for one dimensional array processing: -
 - Declare and initialize an array called **num** with prime numbers 2, 3, 5, 7, 11, 13, 17.
 - Double up the values in the **num** array, total up and display those values which have remainder of 1 after divided by 3.
 - Display the total and the average of these values which have remainder 1 after divided by 3.

Note: Use appropriate loop structure and declare all the necessary variables.

Two Dimensional Array

2. Demonstrate how to store the student information as shown below into two (2) one-dimensional arrays and one (1) two-dimensional array by using C++ code. Briefly explain your answers.

student name	student Index
Anne	x1001
Benson	x1039
Cherry	x1142
Donald	x1092
Edison	x1003

3. Given the following two dimensional array declaration:

```
int x[2][3] = { {14,3,-5}, {0,46,7} };
```

- a) Illustrate the array above in a row-column format.
 - b) Use a *nested-for* loop to prompt 6 contiguous numbers from the user and assign the values to the array **x**.
 - c) Use a nested-for loop to read the values from array **x** and write codes to display the total, average, maximum number and minimum number.
4. Declare a two-dimensional array called **Arr2D** of integer type which has 2 rows and 4 columns. Then write appropriate code segments for the following:
 - a) Use *for* loop to obtain input from the user and store it into the **Arr2D** array.
 - b) Find and display the total of 2nd row of the **Arr2D** array.

- c) Find and display the total of last column of the **Arr2D** array.
 - d) Find the grand total and average of all the values stored in the **Arr2D** array.
5. Given the fruit price list from three different stores as shown below.

Fruit	Store A	Store C	Store T
Orange	0.80	1.00	0.69
Apple	1.20	1.50	1.80
Pear	1.50	1.55	1.29
Peach	2.00	1.99	1.80

- (a) Demonstrate how to store the prices into a two-dimensional array by using C++ code. Briefly explain your answer.
- (b) Write a function named “get_cheap_apple” and use for loop to determine and display the store that sells the cheapest apple.
- (c) Write a function named “storeC_cheapest_fruit” and use for loop to determine and display the cheapest fruit that is sold by Store C.
- (d) Write a function name “get_cheapest_fruit” and use for loop to determine the cheapest fruit amongst all stores. Display the store name.

Vector Array

6. Given an empty vector of integer below:-

```
vector<int> numbers;
```

- (a) Add the values 10, 20, and 30 to the vector
 - (b) Display the size of the vector.
 - (c) Print the first and last element
 - (d) Remove the last element and display the vector again.
- 7.
- (a) Create a vector array to store the fruits banana, apple, orange and cherry
 - (b) Access and display the second fruit in the array
 - (c) Replace orange with blueberry
 - (d) Add a new fruit papaya into the array
 - (e) Display all the fruits

8. You are the secretary of a CEO. You record all the times (24-hour format) of appointment for the CEO every day. All the records are to be stored in vector array.
- a) The CEO will have few appointments today at 10:30, 14:15, 16:00 and 18:30. Record the times of appointments in the array called **appointment**.
 - b) Change the third appointment time to 16:20 as requested by the CEO.
 - c) There is an urgent appointment at 17:10 to meet up with a client. Use a loop to insert it after 16:20.
 - d) However, the first client has called up to cancel the appointment at 10.30. Use a loop to update the record accordingly.
 - e) Display and show the updated appointment times to your CEO.